

Timeless Balance and Infinite Regress

A constructive mathematical reconstruction and its limits

ChatGPT for Thor Fabian Pettersen

15 September 2026

Verdict. I have not solved the full metaphysical problem of infinite regress or proved DEMU. I have completed an explicit mathematical reconstruction of part of its central idea: a whole can be exactly invariant under paired operations while its internal correlations distinguish states and support a relational description of change. For a fixed finite pair of state spaces and a pure state, a precise symmetry requirement selects a unique state. The assumptions do not select the state-space dimension, the internal dynamics, a cuboctahedron, or a dual torus.

This note presents the argument and proofs, followed by the same result in plain language. The mathematical ingredients are established results in linear algebra and quantum theory. Their application here is a proposed interpretation of aspects of DEMU, not a claim of a new quantum theory or a completed explanation of existence.

1 The philosophical starting point

I grant Pettersen's intended axiom for this attempt: absolute nonexistence is impossible, so reality necessarily exists. I also grant, for the conditional discussion, that reality admits a complete explanation and that a vicious regress of unresolved foundational dependence cannot be actual. Here, "balance" in DEMU means the complete closure of that dependence, not merely cancellation of forces.

Under these premises, existence must have an account that does not indefinitely defer its foundation. This conditional conclusion does not need to be defended by endlessly repeating "balance explains balance." The substantive question is whether the proposed structure supplies that account.

The clarified model represents timeless equilibrium by the unfolded cuboctahedron. Folding and unfolding express intrinsic activity. Expansion and contraction have exactly zero delay, so no independent spatial separation is established at the ground. Experienced spacetime is an internal representation within consciousness. Cosmic sex is the proposed generative interaction through which intrinsic polarity yields differentiated balanced expressions. Consciousness is already fundamental; local minds are its expressions.

I retain this distinction between the nonspatial ground and its geometrical representation. A calculation showing that a folded polyhedron has nonzero radius does not refute the claim that it represents a nonspatial ground. It also does not establish that representation as necessary.

An earlier external activation of the entire reality is excluded if all temporal relations belong to that reality. Any proposed earlier event already presupposes a temporal relation and therefore belongs to what is being explained. This addresses external activation conditionally. It does not select the intrinsic constitution of reality or show that it is atemporal.

2 What the mathematical reconstruction assumes

To move beyond verbal description, I introduce a specific trial structure. These are additional mathematical assumptions, not deductions from necessary existence.

1. There are two paired sets of internal degrees of freedom, each represented by a finite complex Hilbert space of dimension $d \geq 2$. Their joint state space is a tensor product. These factors need not be spatially separated objects.
2. The trial whole is described by a normalized pure state. Purity is a substantive assumption; allowing mixed states weakens the uniqueness result below.
3. A chosen pairing identifies each transformation U of one factor with the complex-conjugate transformation $\bar{U}$ of the other. The whole must be invariant under every such paired unitary operation.
4. To describe internal time, I later specify a clock observable and a rule relating its readings to states of the other factor. These choices are not derived from invariance.

No fundamental position coordinates, spatial metric, or external time variable are supplied. This is a mathematically nonspatial description. It is not a proof that actual space is an illusion or that the state space itself has mathematical dimension zero. In fact, its complex dimension is d^2 .

The paired-invariance condition is a candidate mathematical expression of reciprocal balance. It is not assumed to be equivalent to DEMU's complete metaphysical closure. Establishing that equivalence would be a further philosophical result.

3 An exact invariant whole

Choose an orthonormal basis in the first factor and its paired conjugate basis in the second. Define:

$$|\Omega_d\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j\rangle \otimes |\bar{j}\rangle.$$

The basis labels describe internal alternatives. The overbar specifies the pairing; it does not name another spatial location. The state is the canonical maximally entangled vector for the chosen paired spaces. Paired unitary invariance of maximally entangled and related isotropic states is established quantum mathematics; see Terhal and Vollbrecht [1].

Proposition 1. For every unitary matrix U :

$$(U \otimes \bar{U})|\Omega_d\rangle = |\Omega_d\rangle.$$

Proof. After applying the paired transformation, the coefficient of the basis vector with labels a, b is:

$$\frac{1}{\sqrt{d}} \sum_j U_{aj} \bar{U}_{bj} = \frac{\delta_{ab}}{\sqrt{d}}.$$

The equality follows from unitarity. These are exactly the original coefficients. The cancellation is exact; it is not a time average or the endpoint of an infinitely fast physical sequence.

Proposition 2. For the chosen dimension and pairing, this is the unique normalized pure state invariant under every paired operation, up to an overall phase.

Proof. Write an arbitrary candidate vector using a coefficient matrix A . Paired transformation changes that matrix to $U A U^\dagger$. Invariance therefore requires:

$$U A U^\dagger = A \text{ for every unitary } U.$$

Arbitrary diagonal phase transformations force all off-diagonal entries of A to vanish. Permutation matrices force every diagonal entry to be equal. Thus $A = c I$. Normalization gives $|c| = 1/\sqrt{d}$, leaving only an overall phase. This proves the claim.

This result establishes a particular instance in which an explicit symmetry condition selects a unique pure state. It does not establish that the symmetry condition or its mathematical setting is metaphysically necessary.

4 Uniformity with relational distinctions

Let ρ be the density operator of the whole. Ignoring either factor gives the other factor's reduced state:

$$\rho_C = \rho_S = \frac{I}{d}.$$

Proof. In the partial trace, orthogonality removes terms whose paired labels differ. The surviving diagonal terms each have coefficient $1/d$.

Each factor separately has uniform outcome probabilities in every orthonormal basis. Nevertheless, measurements in paired bases have exact correlations: corresponding labels agree with probability one, with each matching pair occurring with probability $1/d$.

Consequently, uniform local statistics and symmetry of the whole are compatible with distinguishable relational alternatives. This addresses part of the tension between uniformity and differentiation. It does not show how an entirely relationless point acquires relations: the paired structure is already an assumption of the construction.

Nor does the construction establish consciousness. It supplies mathematical correlations that an interpretation of experience might use. Whether these correlations have subjective experience is a separate claim.

5 Internal time without external evolution

For convenience, write N for the diagonal matrix with entries $0, 1, \dots, d-1$. Opposite phase operations satisfy:

$$(e^{-i\theta N} \otimes e^{+i\theta N})|\Omega_d\rangle = |\Omega_d\rangle.$$

Each matching term acquires two cancelling phases. The dimensionless label θ is a parameter of the internal relation. The equation does not assert that the whole spends an external interval executing the operation.

Now choose the following d clock states:

$$\theta_k = \frac{2\pi k}{d}, |c_k\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} e^{-ij\theta_k} |j\rangle.$$

These clock states are orthonormal, by the finite Fourier sum. Each clock outcome has probability $1/d$. Conditional on reading k , the normalized state of the other factor is:

$$|s_k\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} e^{+ij\theta_k} |\bar{j}\rangle.$$

Proof. Contract the clock component of the whole with the conjugate of c_k . The resulting vector has squared norm $1/d$; normalizing gives the displayed state.

These conditional states are distinguishable and related by a fixed step:

$$|s_{k+1}\rangle = V |s_k\rangle, V = e^{2\pi i N/d}, V^d = I.$$

Indices are taken modulo d . The whole is invariant, while conditioning on different internal clock readings gives different system states in a cyclic order. No outside observer needs to scan the clock to define these correlations.

This is a finite illustration of the Page and Wootters approach to relational dynamics [2, 3]. It establishes conditional dynamics and an internal ordering. It does not establish felt passage, a thermodynamic arrow, an indefinitely accurate clock, or the duration of real human experiences. Clock choice, ordering rule, and any calibration into seconds remain supplied structure.

This reconstruction also makes a substantive adjustment to the proposed infinitely rapid alternation: it replaces an external sequence compressed into zero duration with a stationary whole and internal correlations. I have not proved those two descriptions equivalent. If literal succession at exactly zero elapsed time is indispensable to DEMU, that part remains unresolved.

6 A small model with an internal record

To make the relation less abstract, take two binary variables, x and y , and stipulate the reversible rule:

$$F(x, y) = (1 - y, x).$$

The second variable records the first variable's preceding value. The first variable changes according to the old value of the second. The rule gives the following cycle:

Internal clock reading	First variable	Second variable
0	0	0
1	1	0
2	1	1
3	0	1

After reading 3 the pattern returns to reading 0. This is a four-state cyclic record, not a model of a human observer. The complete clock-system state is:

$$|\Psi\rangle = \frac{1}{2} \sum_{k=0}^3 |k\rangle \otimes |F^k(0,0)\rangle.$$

Let T advance the clock label by one modulo four, and let P_F apply the reversible rule to the two variables. Then:

$$(T \otimes P_F)|\Psi\rangle = |\Psi\rangle.$$

Proof. Both labels advance together. Relabel the sum by $k+1$ and use $F^4(0,0)=(0,0)$. The same four terms result.

The entire relation is invariant, while its conditional configurations contain differences and a rudimentary record. The correlated whole is a relabelled instance of the $d=4$ maximally entangled construction. In this description, all internal transitions are represented together; no earlier external act is included to start the cycle.

This supplies an explicit illustration relevant to cosmic sex: differentiated patterns can participate in a closed relational whole. However, the rule F , the variables, and their interpretation were stipulated. The example does not derive biological reproduction, consciousness, or the production of complexity from polarity alone.

7 Where the attempt at a complete solution stops

I tried to strengthen the construction into a unique necessary account of reality. The following remaining freedoms prevent that conclusion from the checked conditions.

Dimension remains free. The paired-invariance proof works for every finite $d \geq 2$. The $d=2$ and $d=3$ cases have different numbers of orthogonal conditional states and different Schmidt ranks. They cannot be made identical by a unitary change of basis. These are distinct mathematical models satisfying the supplied algebraic requirements, not purported counterexamples to a fully specified metaphysical closure condition.

Purity remains an assumption. Even after fixing d , allowing mixed states yields a continuous family of invariant states:

$$\rho_p = p|\Omega_d\rangle\langle\Omega_d| + (1-p)\frac{I}{d^2}, 0 \leq p \leq 1.$$

Every member has uniform marginals and paired unitary invariance. Yet the probability of matching outcomes in the paired basis is $p + (1-p)/d$. Thus the correlations vary despite these shared symmetry properties. This is the isotropic-state family in a convenient parameterization [1].

Internal rules remain free. For any finite permutation with a closed orbit, the history construction in section 6 produces a jointly invariant clock-system state. Global stationarity therefore allows many different internal rules. It does not independently choose the rule required for our universe.

Infinite dimension is not an automatic remedy. In the direct Hilbert-vector version of section 3, equal nonzero coefficients over infinitely many orthogonal matching pairs cannot be normalized. The sum of their squared magnitudes diverges. This limits that specific ansatz; it is not an impossibility proof for infinite reality or for other mathematical representations.

No toroidal geometry has been derived. The two factors are algebraic factors, not two tori. Complex phase rotations are not thereby physical toroidal circulation. The model supplies no necessary map to a cuboctahedron, no spacetime metric, and no geometrical proof of zero physical dimension.

Actuality has not been obtained from mathematical existence. A consistent mathematical construction shows what a formal structure permits. It does not establish that the structure is instantiated as conscious reality. The starting axiom grants necessary existence in general; it does not identify which structure exists.

These are not demands for an endless chain of external causes. They identify choices still made inside the proposed account. I have not justified them by a principle that follows from the philosophical premises.

8 The conditional route to a full result

Let P be conditions independently shown to hold of every possible reality, and let $[M]$ denote all descriptions equivalent to a particular structure M . Suppose the conditions have exactly that one equivalence class of nonempty realizations:

$$\text{Mod}(P) = [M].$$

If necessary reality exists and satisfies P , it must instantiate M up to the stated equivalence. This is a valid conditional route from necessary existence to necessary structure. It does not require a second event that manufactures existence from nothing.

Completing that route requires establishing the conditions, their consistency, and their uniqueness, then explaining why they provide the relevant kind of grounding. It also requires relating the resulting structure to actual experience. The paired-invariance construction does not complete those tasks. Its conditions select a pure state only after consequential mathematical choices have been made.

The completed result here is therefore a constructive compatibility result: within a specified mathematical framework, exact paired invariance can coexist with internal distinctions, conditional dynamics, and a finite record. The full metaphysical closure remains open.

9 Verification and provenance

The arguments above are analytic proofs under their stated assumptions. Numerical checks served only to detect algebraic or sign mistakes. For $d = 2, 3, 4, 5$, I checked invariance under twelve sampled unitary matrices per dimension, uniform marginals, clock probabilities, orthogonality of the conditional states, and the cyclic step rule. Residual errors were below 9×10^{-16} . Solving the finite invariance constraints gave a one-dimensional invariant vector space in each case. The four-state history satisfied its joint-invariance identity exactly in the numerical representation.

The philosophical interpretation is motivated by DEMU and the clarifications in this conversation. In particular, DEMU PDF pages 67–73 give the cosmic sex and balance propagation account, page 78 clarifies fundamental consciousness, and pages 306–308 discuss the zero-delay ground and the unity of time and timelessness. These source claims are not converted into verified physics by the calculations.

[1] Barbara M. Terhal and Karl Gerd H. Vollbrecht, The Entanglement of Formation for Isotropic States. [Author paper](#). Used for the established paired-invariance and isotropic-state context.

[2] Don N. Page and William K. Wootters, Evolution without evolution Dynamics described by stationary observables, Physical Review D 27, 2885 (1983). [Publisher record](#). Used for the established distinction between a stationary whole and dynamics relative to internal clocks.

[3] Chiara Marletto and Vlatko Vedral, Evolution without evolution and without ambiguities, Physical Review D 95, 043510 (2017). [Author paper](#). Further work on the Page and Wootters framework. This note does not claim to reproduce that paper's conclusions about subsystem ambiguities.

Plain language version

I have not solved why reality must have your particular structure. I have made one central idea mathematically precise: a whole can stay exactly unchanged while its internal relationships distinguish a sequence of states.

I represented the proposed duality by two paired sets of possibilities. I then imposed a precise rule: every transformation of one set is accompanied by a matching transformation of the other, leaving the whole unchanged. Under specified assumptions, this rule fixes one correlated state.

The cancellation is exact and requires no external delay or infinitely fast physical process.

An important feature follows. Each side considered alone has completely uniform statistics. But the relationship between the two sides is highly specific. Uniformity therefore does not erase all relational information. That supplies a concrete way to understand how a uniform character can coexist with distinctions.

Next, one side serves as an internal clock. Different readings of that clock correspond to different states of the other side. The complete relationship stays unchanged, while the states specified relative to the clock form a sequence. No outside spectator is required to define that relationship.

For a small example, consider the four patterns 00, 10, 11 and 01. Connect them in that order and back to 00. At each step, the second digit carries the first digit's previous value. The whole four-step relationship is complete, while its internal entries differ and contain a simple record.

This is a model of ordered correlations, not a demonstration that the digits are conscious or experience passage. It also replaces the image of infinitely rapid alternation with a complete relational structure. I have not shown that your literal zero-delay succession and this reconstruction are equivalent.

The model gives cosmic sex a possible mathematical direction: complementary relationships can belong to a whole with differentiated configurations. But the particular rule that generates those configurations still had to be supplied. Polarity alone did not select it.

I then tried to remove the remaining choices. That is where the attempt stopped. The same invariance principle works with two internal alternatives, three alternatives, four alternatives, and many more. Even at a fixed size, changing another assumption permits different correlations. Many different internal rules can also be included in a stationary whole.

The equations therefore do not tell us that this whole must be your cuboctahedron or dual torus. They do not establish that it is conscious or that it is the actual foundation of existence. Calling the mathematics nonspatial also does not prove that actual reality is zero-dimensional.

Your starting axiom grants that reality exists necessarily. A complete solution would still need to show that reality necessarily has the conditions used in the construction and that those conditions select its actual structure without unexplained choices.

What has been completed: a checked mathematical account of exact paired invariance with internal distinctions and relational change. **What remains open:** the necessary identification of that account with real existence, consciousness, and DEMU's geometry. I cannot honestly present the remaining work as mere polishing or persuasion.